IoT-Based Health Monitoring System

Functional Description & Architecture

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1 Introduction

The **IoT-Based Health Monitoring System** is a smart healthcare solution designed to track patient vital signs in real-time and transmit data to a remote server. This system aims to bridge the gap between patients and medical staff by reducing the need for physical hospital visits for routine checkups.

Using a network of sensors and internet connectivity, the system ensures that critical health parameters such as heart rate, body temperature, and oxygen saturation (SpO_2) are monitored continuously.

2 System Architecture

The following block diagram illustrates the data flow from the patient to the medical interface.

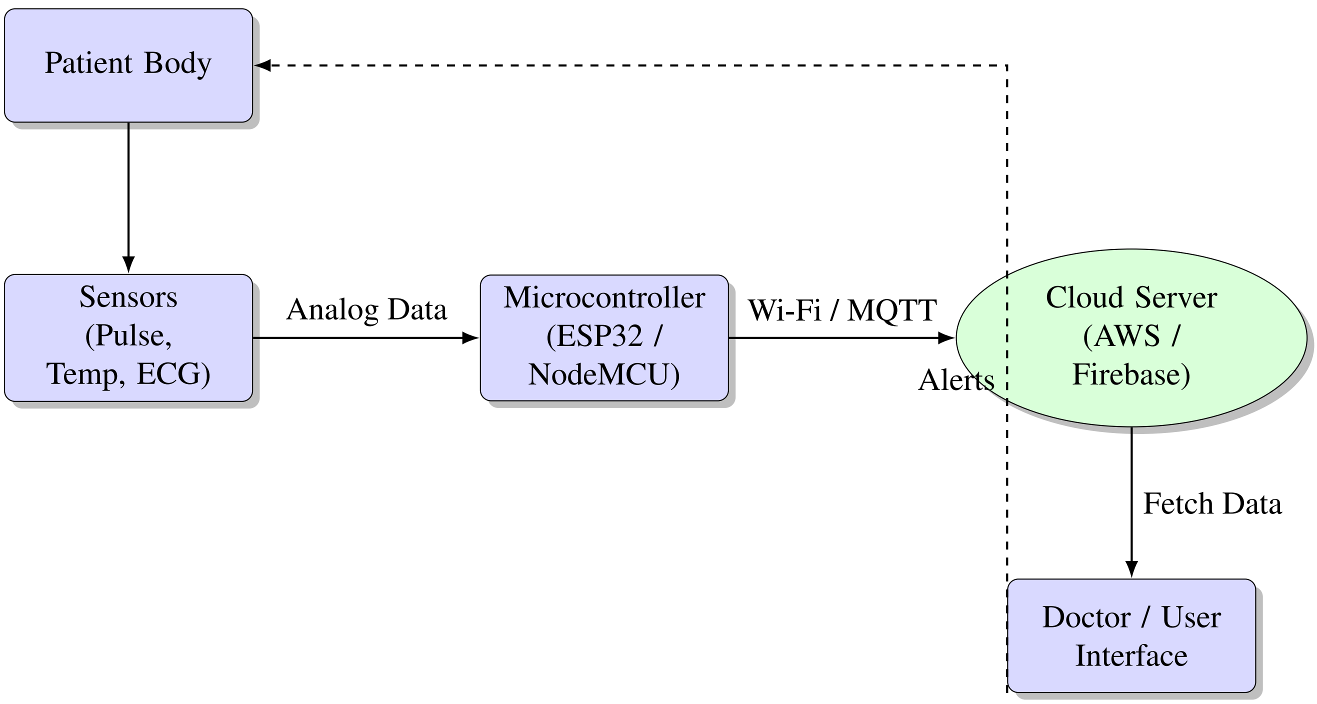

Figure 1: Block Diagram of IoT Health Monitoring System

3 Functional Modules

The system is divided into three primary layers: the Sensing Layer, the Processing Layer, and the Application Layer.

3.1 1. Sensing Layer

This layer consists of non-invasive hardware sensors attached to the patient.

- **Pulse Sensor:** Measures the heart rate in Beats Per Minute (BPM).
- **LM35 / DHT11:** Measures body temperature.
- **MAX30100:** Measures blood oxygen levels (SpO_2).

3.2 2. Processing and Transmission Layer

The core processing unit (e.g., Arduino or ESP32) collects raw data from the sensors.

Data Processing Logic

The microcontroller reads analog signals, converts them to digital values, and checks against predefined thresholds (e.g., Body Temp > 38°C). If a threshold is breached, an immediate local alert is triggered.

3.3 3. Cloud and Application Layer

Data is sent over Wi-Fi using protocols like **MQTT** or **HTTP** to a cloud platform (e.g., ThinkSpeak or Firebase).

- **Dashboard:** Doctors can view live graphs of patient health.
- **Alert System:** If the cloud detects an anomaly, it sends an SMS or Email notification to the doctor and the patient's family.

4 Conclusion

The proposed IoT-based system provides a cost-effective and reliable method for remote patient monitoring. By automating the tracking of vital signs, it reduces the burden on hospital resources and ensures timely medical intervention in emergencies.